EECE 5554 Lab 4: Navigation with IMU and Magnetometer

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GitLab: gitlab.com/shastriNi/eece-5554 (LAB4: src, data, analysis, report)

1. Introduction

In this lab I used a VectorNav VN-100 IMU and a GPS receiver to estimate the heading, forward velocity and trajectory of a car. The recorded drive was split into two parts: the circles, which were used only to calibrate the magnetometer, and the route after the circles, which was used for everything else. Heading was estimated from the magnetometer and the gyroscope and combined with a complementary filter, velocity was estimated by integrating the forward acceleration, and these were combined to get the trajectory by dead reckoning. GPS was only used as a reference to compare against; all estimates were computed from the IMU data only. The analysis was done in Python using numpy and matplotlib. The bag file was read with the mcap and mcap-ros2-support libraries. The low-pass filter (first-order Butterworth, applied forward and backward) and the trapezoidal integration were written in numpy, because the scipy library could not run on my laptop. They give the same result as scipy filtfilt and cumulative_trapezoid.

2. Part 1: Driver and Stationary Noise Analysis

The driver from the previous lab was reused (LAB4/src, imu_driver/driver.py). It opens the serial port at 115200 baud and sets the output by writing to the IMU registers: $VNWRG,06,14 sets the output type to VNYMR and $VNWRG,07,40 sets the rate to 40 Hz. Each $VNYMR line is checked with its checksum (XOR of all characters between $ and *), and lines with a wrong checksum are skipped. Valid lines are split into yaw, pitch, roll, magnetometer, accelerometer and gyro values. Yaw, pitch and roll are converted from degrees to radians and then to a quaternion (ZYX order) without libraries, and the magnetometer is converted from Gauss to Tesla. The data is published on the topic imu as a custom message containing a header (frame_id IMU1_Frame), sensor_msgs/Imu, sensor_msgs/MagneticField and the raw string. The serial port is passed through the launch file, for example ros2 launch imu_driver imu_launch.py port:=/dev/ttyUSB0.

About 18 minutes of data was recorded with the IMU stationary. The IMU was upside down, so roll is about 177 deg and accel Z is about +10 m/s². The start and end, where the IMU was being placed and picked up, were found using the gyro magnitude (above 0.01 rad/s) and removed, leaving 980 s (39,201 samples). Figure 1 shows the time series and Table 1 the mean and standard deviation of each channel.

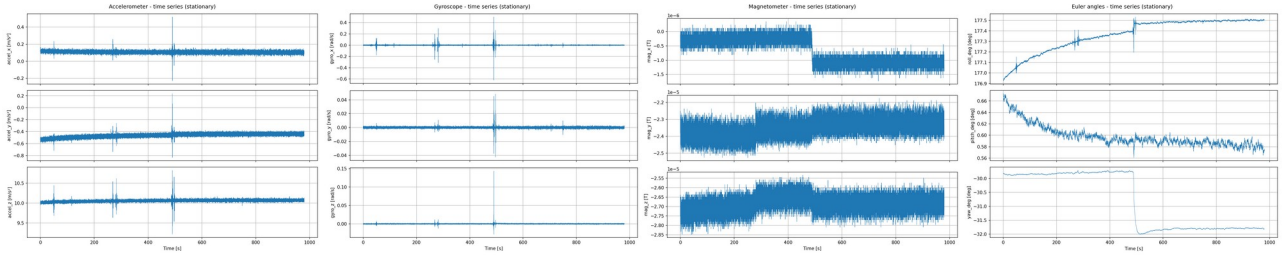


Figure 1. Time series of the stationary data: accelerometer, gyroscope, magnetometer and Euler angles (left to right).

Table 1. Mean and standard deviation of the stationary data.

Signal	Mean	Std. dev.	Signal	Mean	Std. dev.
Accel X (m/s²)	0.1066	0.0153	Mag X (µT)	-0.677	0.447
Accel Y (m/s²)	-0.4619	0.0341	Mag Y (µT)	-23.49	0.545
Accel Z (m/s²)	10.0598	0.0300	Mag Z (µT)	-26.86	0.449
Gyro X (rad/s)	-3.40e-5	8.73e-3	Roll (deg)	177.366	0.150
Gyro Y (rad/s)	7.63e-5	1.10e-3	Pitch (deg)	0.598	0.019
Gyro Z (rad/s)	1.02e-5	1.28e-3	Yaw (deg)	-30.807	1.006

The data has a few bumps at about 50 s, 270 to 285 s and 490 s, mostly about the X axis. Roll and pitch drift slowly during the first 490 s, and accel Y changes by about 0.1 m/s² over the same time, which matches the roll change (g sin(0.57 deg) is about 0.1 m/s²), so the IMU was tilting slightly. The bump at 490 s rotated the IMU, which caused a 2 deg step in yaw and steps in the magnetometer. The gyro means are close to zero, so the gyro bias is small. This matches Lab 4, where the gyro z bias measured in the parked car was also almost zero. The accelerometer magnitude

is 10.07 m/s², slightly more than g, because of bias and scale errors. The magnetic field magnitude is about 35.7 µT, lower than the expected Earth field of about 50 µT, because the magnetometer was not calibrated and indoor fields are distorted by nearby metal and electronics. Gyro X has a much larger standard deviation than Y and Z because of the bumps. So the standard deviations include both the sensor noise and the extra variation from the bumps and the slow tilt. A quieter location would give values closer to the pure sensor noise.

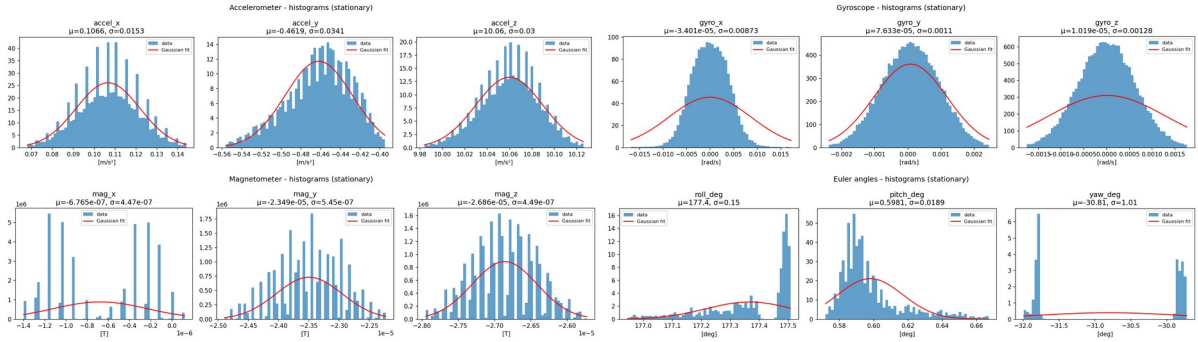


Figure 2. Histograms of the stationary data (0.5 to 99.5 percentile range) with a Gaussian fit using the sample mean and standard deviation.

The histograms (Figure 2) show the noise distribution of each channel. All three accelerometer axes follow a Gaussian distribution well, as expected for sensor noise that comes from many small independent sources. The comb pattern in the bars is from the 0.001 m/s² output resolution. The gyroscope is also bell-shaped around zero, but X and Z are more peaked than the Gaussian fit, because the few large bump values increase the standard deviation. Gyro Y, which had the fewest disturbances, fits the Gaussian well. The magnetometer histograms have separate spikes because its output is quantized. Y and Z still roughly follow a Gaussian shape, while X has two groups because of the shift at 490 s. The Euler angles are not Gaussian: roll has a long tail from the slow drift, pitch is skewed, and yaw has two peaks from the step at 490 s. These shapes come from the IMU moving slightly, not from random noise. In a quiet 56 s part of the data, roll and pitch had standard deviations of only 0.004 and 0.006 deg. Overall, the accelerometer and gyro noise is roughly Gaussian with a small bias, and most of the non-Gaussian behavior comes from the environment, not the sensor.

3. Part 2a: Data Collection and Splitting the Data

The IMU was taped flat in the car with the x-axis pointing forward, and the GPS puck was placed on the roof. Both sensors were recorded in one ROS 2 bag: /imu at 40.1 Hz and /gps at 1 Hz, for a total of 1254 s. The car was parked for about 15 s, drove about 2.7 circles for the compass calibration, and then drove a 2.3 km loop through Boston that ended about 22 m from the end of the circles. The VN-100 frame is x forward, y right and z down, so a negative gyro z is a left turn and yaw increases clockwise from north.

Since the circles are only a calibration maneuver and not part of the route, the data was split into two files (Figure 3). circles.npz contains only the circles (25 to 88 s, 2520 IMU samples) and was used only for the magnetometer calibration. route.npz contains everything after the circles (from 88 s to the end, 46,644 IMU samples and 1167 GPS samples, 1166 s) and was used for all other parts. Time and the GPS east/north position were set to zero at the start of the route, so the route starts exactly where the circles end.

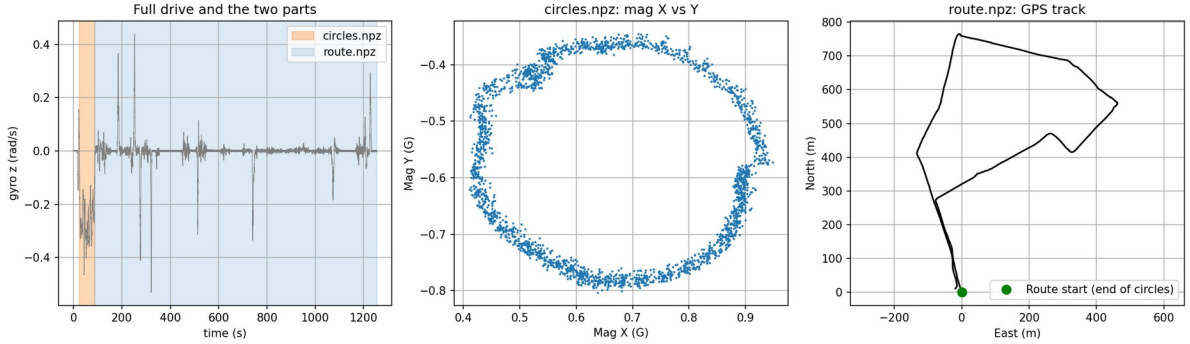


Figure 3. Gyro z of the full drive with the two parts marked (left), magnetometer data of the circles (middle) and GPS track of the route (right).

3.1 Stops, Biases and Starting Speed

Stops were found using only the IMU. Over a 1 s window, the car is stopped if gyro z is close to zero (|mean| < 0.02 rad/s, not turning) and accel x does not change much (standard deviation < 0.05 m/s², since a stopped car vibrates much less than a moving one), for at least 2 s. This found 12 stops with a total of 337 s (28.9% of the route). The gyro z bias was taken as the mean over all stops (-4.2e-5 rad/s, almost zero, which agrees with the Lab 3 stationary data). The accel x bias was taken from the first stop (0.421 m/s²). Since the route starts while the car is still moving at the end of the circles, the starting speed was also needed. It was found from the lateral equation in Section 6 ($y_{obs} = \omega \dot{X}$), which gives $\dot{X} = y_{obs} / \omega$ during the steady turning of the circles. Using the last 10 s of the circles gave a starting speed of 3.42 m/s, which is close to the GPS speed of 3.1 m/s at that time.

4. Part 2b: Heading Estimation

4.1 Magnetometer Calibration

The magnetometer inside the car is affected by hard-iron and soft-iron effects. Hard iron is a constant field from magnetized parts of the car, which shifts the circle of X-Y readings away from the origin. Soft iron is caused by nearby metal that bends the Earth's field, which changes the circle into a tilted ellipse. Both were corrected using circles.npz, since the car faced every direction during the circles. An ellipse $Ax^2 + Bxy + Cy^2 + Dx + Ey = 1$ was fitted to the X-Y data with least squares. The center of the ellipse is the hard-iron offset, which was subtracted. The eigenvectors and eigenvalues of the ellipse give the directions and lengths of its axes, and the soft-iron matrix S rotates the data onto these axes, scales both axes to the average radius and rotates back. The correction m_corrected = S (m - offset) was then applied to all route samples. The results are given in Table 2.

Quantity	Value
Hard-iron offset	(0.671, -0.571) G
Ellipse semi-axes	0.246 G and 0.210 G
Radius after correction	0.228 G
Soft-iron matrix S	[[0.926, -0.006], [-0.006, 1.087]]

Table 2. Magnetometer calibration results.

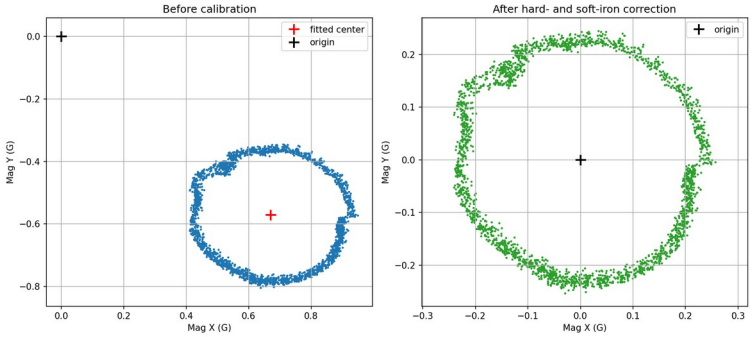


Figure 4. Magnetometer X-Y data during the circles before (left) and after (right) calibration.

Before the calibration, the center of the data was 0.88 G from the origin, which is more than the radius of the circle (0.228 G). The origin was outside the data, so the heading from the raw data would never complete a full turn, and the raw magnetometer cannot be used for heading in this car. After the calibration the data is a circle centered at the origin. S is almost diagonal, so the soft-iron effect is mostly a stretch along the sensor axes.

4.2 Yaw from Magnetometer and Gyroscope

The magnetometer yaw was calculated from the corrected route data as yaw = atan2(-m_y, m_x), where the minus sign is needed because the y-axis points to the right, so this gives the heading clockwise from magnetic north. The gyro yaw was found by integrating gyro z with the trapezoidal rule after subtracting the bias, starting from the first magnetometer yaw (4.5 deg, almost north, which matches the first part of the route). The bias has to come from a fully stationary period. In an earlier attempt the bias window included the start of a turn, which gave a wrong bias of -0.006 rad/s and a fake drift of about 0.34 deg/s, so the mean over all detected stops was used instead.

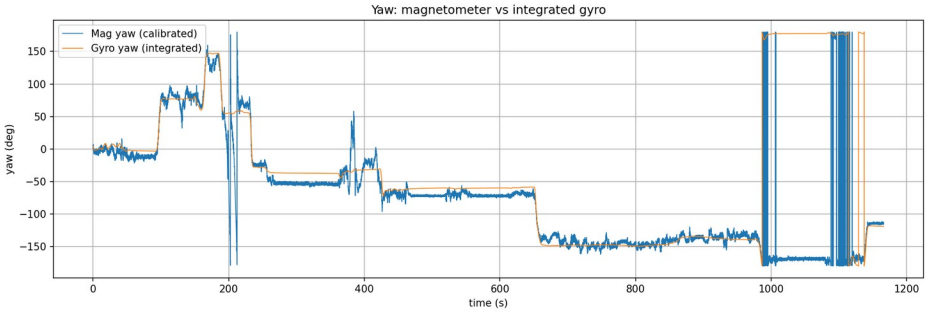


Figure 5. Yaw from the calibrated magnetometer and from the integrated gyroscope on the route (wrapped to ±180 deg).

Both methods show the same turns at the same times (Figure 5). The vertical lines are where the angle wraps between +180 and -180 deg, which happens a lot after 985 s because the car is driving south. The gyro yaw is smooth and not affected by magnetic disturbances, but it drifts slowly, and between about 250 and 650 s it is 10 to 18 deg above the magnetometer yaw. The magnetometer yaw does not drift, but it is noisy and has large errors at about 200 to 215 s, where it drops to -180 deg and back while the gyro shows a steady heading, and at about 370 to 410 s, where it jumps up to +60 deg while the car is driving straight. This is most likely caused by nearby metal such as buildings, vehicles or power lines. So the gyro is better in the short term and the magnetometer is better in the long term.

4.3 Complementary Filter

The complementary filter combines the low-frequency part of the magnetometer yaw, which does not drift, with the high-frequency part of the gyro yaw, which is smooth. Since a high-pass filter is one minus the low-pass filter, LPF(mag) + HPF(gyro) can be written as yaw = gyro + LPF(mag - gyro). This form was used, with the difference wrapped to ±180 deg before filtering. The filter is a first-order Butterworth low-pass filter with a cutoff of 0.05 Hz, applied forward and backward so there is no time delay, so the magnetometer is trusted for changes slower than about 20 s and the gyro for faster changes. Wrapping the difference is important: if the magnetometer yaw is unwrapped with np.unwrap and then filtered, the disturbances cause jumps larger than 180 deg between samples, which unwrap treats as full turns. Table 3 shows that this adds about two extra turns.

Signal	**Total yaw change over the route**
Gyroscope (integrated)	-123 deg
Complementary filter	-118 deg
VN-100 yaw	-126 deg
Magnetometer (np.unwrap only)	-840 deg (about 2 extra turns)

Table 3. Total change in unwrapped yaw for each method.

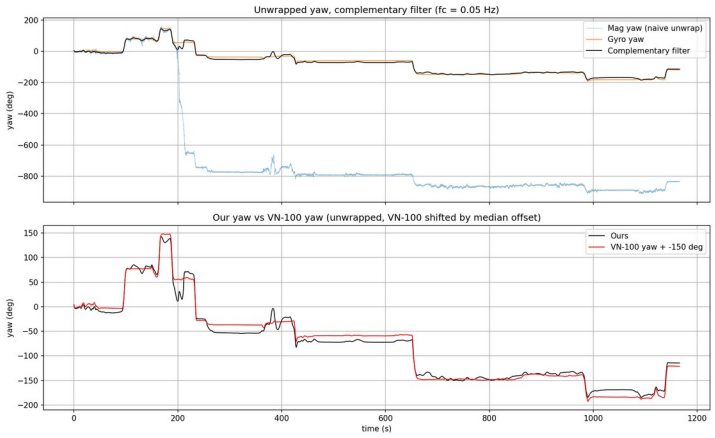


Figure 6. Top: unwrapped magnetometer, gyro and complementary filter yaw. Bottom: complementary filter yaw and VN-100 yaw shifted by the median offset of -150 deg.

The filtered yaw is as smooth as the gyro yaw but follows the magnetometer over long times. Compared with the yaw from the VN-100, all turns happen at the same time and are the same size, and the total change is almost the same (-118 vs -126 deg). There is an offset of about -150 deg (median) because the VN-100 calculates its yaw from its own magnetometer, which is not corrected for the hard-iron and soft-iron effects of this car (Section 4.1). In some parts (about 260 to 360 s, 430 to 650 s and 1000 to 1130 s) the two differ by 10 to 15 deg, because the uncorrected VN-100

magnetometer has an error that depends on the direction, and our heading also has some error. At about 200 to 215 s and 380 s the longer magnetic disturbances partly pass through the 0.05 Hz filter.

5. Part 2b: Forward Velocity

The forward velocity was found by integrating accel x. The GPS velocity was calculated as the distance between consecutive UTM positions divided by the time between them, and one point that gave 31 m/s (a GPS position jump) was removed. The velocity from integrating the raw acceleration does not make sense: it reaches 320 m/s at the end of the route (Figure 7, top), because the constant bias of 0.42 m/s² (sensor bias plus a small tilt that adds part of gravity to the x-axis) becomes a ramp when integrated. The following adjustments were made:

- **Removing the bias:** the accel x value at the first stop (0.421 m/s²) was subtracted, because the real acceleration is zero when the car is stopped.
- **Zero velocity at stops:** the velocity was set to zero at all 12 stops found in Section 3.1.
- **Removing the bias in each segment:** between two stops the velocity has to start and end at zero. If the integral does not end at zero, the bias changed (road slope, tilt while speeding up and braking), so the constant bias that makes it end at zero was removed, which is the same as subtracting a straight line from the velocity in that segment.
- **Starting speed:** the first segment starts at the starting speed of 3.42 m/s from Section 3.1 instead of zero, and still has to end at zero at the first stop.
- **No negative speed:** the car never drove backwards, so negative values were set to zero.

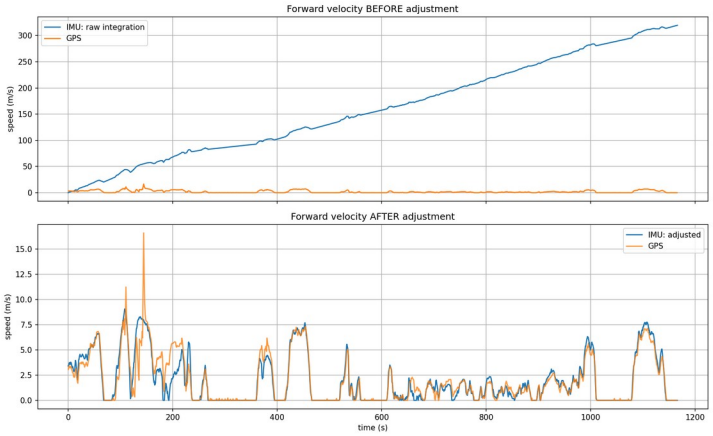


Figure 7. Forward velocity from the IMU and GPS before (top) and after (bottom) the adjustments.

After the adjustments, the IMU velocity matches GPS well, with an RMS difference of 0.86 m/s. The first part (0 to 70 s) starts at 3.4 m/s and follows GPS up to 6.8 m/s and back to the first stop (RMS 0.56 m/s), and from about 250 s to the end the peaks, the stops and the stop-and-go traffic all match (RMS 0.42 m/s). Between about 100 and 250 s the IMU is sometimes lower than GPS (RMS 2.1 m/s), but the GPS is not reliable here either, since it has spikes of 11 and 16.6 m/s that are not realistic on this road. This is the same area where the magnetometer had disturbances, so tall buildings may have caused GPS multipath errors.

6. Part 2b: Dead Reckoning

If the car moves in a plane, the acceleration measured by the IMU at position (x_c, 0) in the vehicle frame is $x_{obs} = \ddot{X} - \omega\dot{Y} - \omega^2 x_c$ and $y_{obs} = \ddot{Y} + \omega\dot{X} + \dot{\omega}x_c$, where X is the position of the center of mass and ω is the yaw rate. If the car does not slide sideways ($\dot{Y} = 0$) and $x_c = 0$, then $\ddot{X} = x_{obs}$, which was integrated to get $\dot{X}$ in Section 5, and $y_{obs} = \omega\dot{X}$.

6.1 Comparing $\omega\dot{X}$ and y_obs

$\omega\dot{X}$ was calculated by multiplying gyro z by the adjusted velocity and compared with accel y, with both smoothed by a 1 Hz low-pass filter. Accel y reads between -0.43 and -0.64 m/s² at most stops, which is part of gravity due to the road slope and tilt, not real acceleration. This offset was measured at each stop, interpolated between the stops and subtracted.

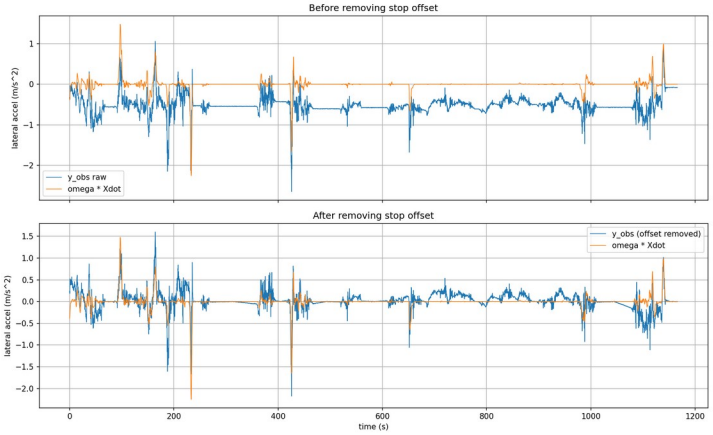


Figure 8. Lateral acceleration y_obs and ωẊ before (top) and after (bottom) removing the offset of accel y.

After removing the offset, the peaks during turns match in time, sign and size, for example about -1.6 and -2.2 m/s² at 232 s and both about +1.0 m/s² at 1139 s, and the correlation while moving is 0.64. In the first 60 s y_obs is about 0.4 m/s² higher than ωẊ, because the route starts while moving and the offset there is taken from the first stop at 69 s, where the road slope is different. The same happens around 1080 to 1140 s. The other differences come from vibration, the roll of the car and the ωẋ_c term, which was ignored here.

6.2 Trajectory

The complementary filter yaw was changed from magnetic to true north by adding the magnetic declination in Boston (-14 deg). The velocity was split into east and north components with v_e = v sin(ψ) and v_n = v cos(ψ) and integrated to get the position. Both tracks start at (0, 0), which is where the circles end. To match the heading, the first straight part of the route was used (20 to 30 s, where the speed is above 3 m/s and the GPS direction changes by less than 10 deg), and the IMU track was rotated so that its direction in this part matches the GPS direction. The scale factor is the GPS path length divided by the IMU path length.

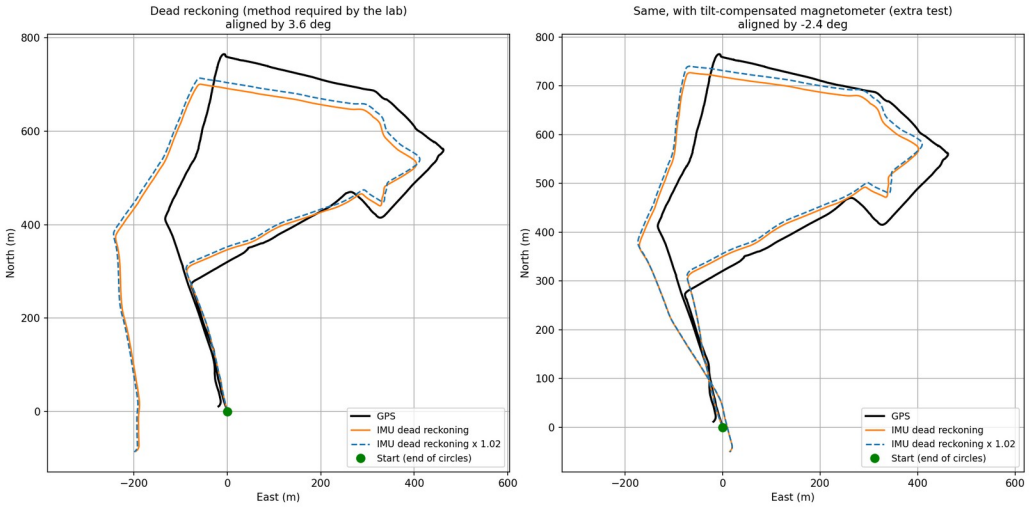


Figure 9. IMU dead reckoning compared with GPS. Left: method required by the lab. Right: extra test with a tilt-compensated magnetometer.

Quantity	Required method	With tilt compensation (extra)
Rotation to align the heading	3.6 deg	-2.4 deg
Path length: GPS / IMU	2308 m / 2266 m	2308 m / 2266 m
Scale factor (GPS / IMU)	1.02	1.02
Average position error	88 m	54 m
Distance between end points	201 m	70 m

Table 4. Trajectory comparison with GPS.

With the required method (Figure 9, left), the heading only had to be rotated by 3.6 deg and the scale factor is 1.02, so the IMU path length is almost the same as GPS. The first part going north and the long part to the east-northeast

overlap the GPS track closely. After that the IMU track slowly turns away from GPS: the upper loop is smaller and the last part going south ends about 200 m west of the GPS end point. Since the path length is correct, this error mostly comes from the heading. Small heading errors of a few degrees in each part add up, because both the heading and the velocity are integrated into the position and there is nothing to correct them.

Starting the route at the end of the circles also made a big difference. An earlier version used the whole drive including the circles, where the velocity was much too low because the car was turning all the time, and this gave a scale factor of 1.40. Splitting the data and using the starting speed from y_obs / ω fixed this.

Tilt compensation (extra test). To check where the remaining heading error comes from, an extra test was done that is not part of the required method. The IMU is tilted by about 1.8 deg and the magnetic field in Boston points steeply downward (mag Z is about 0.87 G compared with 0.22 G horizontally), so even a small tilt adds part of the vertical field to mag X and Y and causes a heading error that depends on the direction. In the test, the magnetometer data was rotated to level using the VN-100 pitch and roll before the calibration (offset (0.704, -0.586) G instead of (0.671, -0.571) G). This reduced the average position error from 88 m to 54 m and the distance between the end points from 201 m to 70 m (Figure 9, right), and the track then almost closes the loop. Since tilt compensation is not part of the lab requirements, the required method was kept for the results above, but this shows that tilt is the main remaining source of heading error.

6.3 Estimating x_c (Optional)

From the y equation with $\ddot{Y} = 0$, $y_{obs} - \omega\dot{X} = \dot{\omega}x_c$, so x_c is the slope between the residual $y_{obs} - \omega\dot{X}$ and $\dot{\omega}$, found with least squares. $\dot{\omega}$ was found by differentiating the 1 Hz low-pass filtered gyro z. A first fit using all moving data gave x_c of about 2.0 m but a weak fit ($R^2 = 0.09$), because the residual also contains slow changes from road slope and roll. To remove these, the residual and $\dot{\omega}$ were high-pass filtered at 0.05 Hz, and only samples where the car was turning ($|\omega| > 0.05$ rad/s, 2372 samples) were used, since $\dot{\omega}$ is almost zero on straight roads.

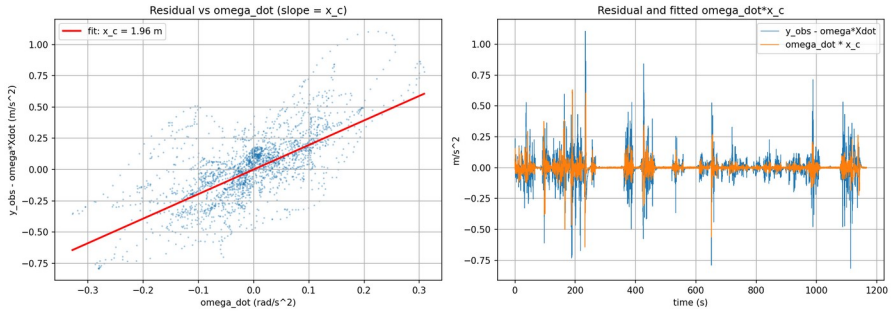


Figure 10. Residual $y_{obs} - \omega\dot{X}$ against $\dot{\omega}$ with the fitted line (left) and over time (right).

This gave $x_c = 1.96$ m with $R^2 = 0.47$, and the result stayed between 1.9 and 2.3 m for different filter settings, so the estimate is consistent. The positive sign means the IMU is in front of the center of mass. It is still a rough estimate, since $\dot{\omega}$ is noisy and the residual also includes roll and vibration.

7. Conclusion

- The raw magnetometer cannot be used for heading in a car. After hard-iron and soft-iron calibration it does not drift, but it is noisy and affected by nearby metal.
- The gyro yaw is smooth and not affected by magnetic disturbances, but it drifts, and the bias has to be measured when the car is fully stationary.
- The complementary filter combines both and matches the VN-100 yaw in shape and total rotation (-118 vs -126 deg). Filtering the wrapped difference avoids the extra turns caused by unwrapping the disturbed magnetometer yaw.
- The integrated acceleration cannot be used without corrections. After removing the bias, setting the velocity to zero at stops, removing the bias in each segment and using the starting speed from y_obs / ω, it matches GPS with an RMS difference of 0.86 m/s.
- Separating the calibration circles from the route improved the dead reckoning a lot (scale factor 1.02 instead of 1.40). The remaining error mostly comes from heading and keeps growing with distance, so the IMU alone cannot be used for long distances. GPS is needed to limit the error, while the IMU gives smooth, high-rate data between GPS readings. Tilt compensation of the magnetometer could reduce the remaining error further.